

EDUCATION

Qualification	Institution	Year	Score
Graduation	Dhirubhai Ambani University (DAIICT)	2023–27	6.83
12th(CBSE)	Nalanda Vidhyalaya	2022–23	89.2%
10th(CBSE)	Aditya Birla Public School	2020–21	80.8%

AWARDS AND ACHIEVEMENTS

- **Pupil** @Codeforces(Rating: 1229) and **Expert** @Kaggle (Global rank: 285).
- Achieved a top rank in a Kaggle Hedge Fund Competition by building a multivariate two-way model for financial panel data forecasting.
- Conducted in-depth research on the classical puzzle **Instant Insanity** and its mathematical variations.
- Completed independent project on **Submodularity** for Machine Learning and sampling algorithms.
- Achieved **CRL 12,111 (98.98 PR)** in JEE Mains 2023.

PROJECTS

- **Quantitative Derivatives Pricing & Risk Engine** (*FastAPI, NumPy, Numba, Redis, PostgreSQL*) 
 - Engineered an institutional-grade derivatives pricing engine supporting **Black-Scholes**, **Heston Stochastic Volatility**, **Merton Jump-Diffusion**, and CRR Binomial Trees with **Numba JIT compilation** achieving C-level execution speeds.
 - Built a **CVA (Credit Value Adjustment)** engine computing Expected Exposure profiles via Monte Carlo path simulation, and a **Student-t Copula** risk engine for modeling heavy-tailed cross-asset dependencies.
 - Developed a full **Order Management System** with pre-trade risk checks, live broker integration (Zerodha Kite API), and real-time market data streaming via WebSockets.
 - Deployed as a production platform on **Vercel + Render + Upstash Redis**, with JWT authentication, Redis-backed Bloom Filters for $\mathcal{O}(1)$ duplicate detection, and a distributed caching layer for sub-millisecond pricing responses.
- **Wealth Management Engine** (*Python, PostgreSQL, psycpg2*) 
 - Built a Python-driven portfolio management engine supporting real-time NAV ingestion, dynamic holdings tracking, and accurate return computation (CAGR, XIRR).
 - Designed portfolio analytics modules for asset allocation, performance attribution, and rebalancing logic.
 - Developed robust data pipelines using psycpg2 with triggers, constraints, and automated updates to ensure consistency and scalability for multi-user portfolios.
- **Real-Time Trading & Risk Monitoring System** (*Java, MySQL, System Design*) 
 - Engineered a low-latency trading signal system with real-time market data ingestion (1-second intervals) and automated BUY/SELL recommendations.
 - Designed arbitrage detection logic across exchanges, incorporating moving-average strategies for actionable trade insights.
 - Developed a modular architecture for extensible financial instruments and risk monitoring, demonstrating production-style system design.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, SQL, TypeScript/JavaScript
- **AI/ML & Data Science:** TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face, Pandas, NumPy, SciPy
- **Backend & Web Frameworks:** FastAPI, Next.js, React, Celery
- **Databases & Infrastructure:** PostgreSQL (Triggers, Query Optimization), MySQL, Redis, Git, Docker, Render, Vercel
- **Core Competencies:** Quantitative Finance, System Design, NLP, Deep Learning, Competitive Programming
- **Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Object-Oriented Programming, Parallel & Distributed Algorithms, Machine Learning, Big Data Processing

POR & EXTRA-CURRICULARS

- **Mentor & Event Organizer (Khelaiya Club, DA-IICT):** Led workshops and mentored junior students, while organizing various cultural events and club activities. (Jul'24 – Present)
- **NGO Volunteer (Agastya International Foundation):** Assessed impact of hands-on science learning programs on school students by conducting interviews with 10+ students. Analyzed responses to evaluate changes in curiosity and problem-solving skills, developing 5 detailed case studies showcasing student growth. (Dec'24)